

ILVI project: Results and statistics

Introduction

Summary of ILVI study in vivo data showing all statistical computations. Comprehensive for all ILVI project animals studied in the 2021-22 and 2022-23 seasons. This version created in spring 2025 to accompany S. Jonker data deposition into Figshare. Questions about this document can be directed to M. Hagen: mhagen2@fredhutch.org.

This document contains tabular and statistical summaries for:

- Endpoint morphometry
- Chronic study:
 - Blood contents and characteristics
 - Hemodynamics
- Terminal study echo:
 - Myocardial strain (STE / Tomtec)
 - Chamber geometry & EF (STE /Tomtec)
 - Myocardial perfusion (CEUS)
- Other terminal study data:
 - Pre and post ABGs
 - Hemodynamics recorded before echo
 - Hemodynamics recorded during CEUS studies

```
library(emmeans)
library(lubridate)
library(nlme)
library(tidyverse)
library(readxl)
library(here)
library(knitr)

pstar <- function(x){
  if(x<0.001) {"***"}else{
    if(x<0.01){**"}else{
      if(x<0.05){*"}else{
        if(x<0.10){"."}else{
          " "
        }
      }
    }
  }
}

load(here("figshare/ILVI.rda"))
```

Sample sizes and basic information:

```
weights |>
  filter(Allocation=="OHSU",Weights==TRUE) |>
  group_by(Group, Sex) |>
  summarise(N = n()) |>
  pivot_wider(names_from = Sex, values_from = N) |>
  kable()
```

Group	F	M
Inflate	1	6
Sham	6	3

```
# Are sexes statistically different between group per Fisher's exact test? #
```

```
sextable=matrix(ncol=2,nrow=2)
sextable[,1]=c(1,6);sextable[,2]=c(6,3)
fisher.test(sextable)
```

```
##
## Fisher's Exact Test for Count Data
##
## data: sextable
## p-value = 0.06014
## alternative hypothesis: true odds ratio is not equal to 1
## 95 percent confidence interval:
## 0.00156685 1.40629603
## sample estimates:
## odds ratio
## 0.09974122
```

Chronic study

Blood contents and characteristics

All metrics analyzed as two-way RMA treating time as a categorical factor. Where appropriate, the following five post-hoc comparisons are examined, using a Sidak correction: - d0 v d8 within group - d0, d4, d8 between groups

```
abgs |>
  filter(Allocation=="OHSU",ABGs==TRUE, Study_day%in%(c(0,4,8))) |>
  group_by(ID,Sex,Group, Study_day) |>
  select(pH,sO2,pCO2,pO2,Hb,Hct,CtO2,Glucose,Lactate) |>
  pivot_longer(cols=c(5:13), names_to = "Metric", values_to = "Value") |>
  ungroup() |>
  group_by(Group,Metric, Study_day) |>
  summarise(Mean = mean(Value), SD = sd(Value)) |>
  pivot_wider(names_from = Group, values_from = c(Mean,SD)) |>
  relocate(Metric, Study_day,Mean_Sham,SD_Sham, Mean_Inflate,SD_Inflate) |>
  kable(digits = c(0,0,2,2,2,2), caption = "ABG d0,4,8 summary")
```

Table 2: ABG d0,4,8 summary

Metric	Study_day	Mean_Sham	SD_Sham	Mean_Inflate	SD_Inflate
CtO2	0	8.50	0.96	6.66	2.13
CtO2	4	9.43	1.12	6.90	1.76
CtO2	8	8.59	1.32	5.94	1.64
Glucose	0	1.24	0.20	1.26	0.29
Glucose	4	1.21	0.20	1.19	0.27
Glucose	8	1.00	0.26	1.01	0.27
Hct	0	32.11	2.76	35.00	5.47
Hct	4	34.00	4.33	35.79	4.72
Hct	8	36.11	6.25	37.07	4.46
Hb	0	10.22	1.04	10.84	1.53
Hb	4	10.96	1.50	11.43	1.47
Hb	8	11.88	2.86	11.47	1.39
Lactate	0	1.57	0.30	2.13	0.75
Lactate	4	1.51	0.34	2.24	0.60
Lactate	8	1.69	0.40	2.53	0.34
pCO2	0	50.32	3.35	52.24	3.47
pCO2	4	49.18	2.47	53.40	3.68
pCO2	8	50.08	2.23	55.69	2.37
pH	0	7.34	0.06	7.36	0.02
pH	4	7.34	0.03	7.35	0.02
pH	8	7.36	0.02	7.34	0.02
pO2	0	24.18	2.56	19.73	4.93
pO2	4	24.83	6.62	19.17	3.76
pO2	8	24.91	2.13	17.57	2.93
sO2	0	57.12	8.71	50.59	18.83
sO2	4	57.66	12.84	45.29	10.16
sO2	8	61.92	8.43	46.59	14.20

```

abgs |>
  filter(ABGs==TRUE, Study_day%in%(c(0,4,8))) |>
  mutate(Study_day = as.character(Study_day)) |>
  group_by(ID,Sex,Group, Study_day) |>
  select(pH,sO2,pCO2,pO2,Hb,Hct,CtO2,Glucose,Lactate) |>
  pivot_longer(cols=c(5:13), names_to = "Metric", values_to = "Value") |>
  ungroup() |>
  nest_by(Metric) |>
  mutate(mod = list(aov(Value ~ Group * Study_day + Error(ID), data = data))) |>
  summarise(`Group p-value` = summary(mod)[[1]][[1]][1,5], `*` = pstar(`Group p-value`),
            `Day p-value` = summary(mod)[[2]][[1]][1,5], `*` = pstar(`Day p-value`),
            `Interaction p-value` = (summary(mod)[[2]][[1]][2,5]),
            `*` = pstar(`Interaction p-value`)) |>
  relocate(Metric,`Group p-value`,`*`,`Day p-value`,`*`,`Interaction p-value`,`*`) |>
  kable(digits = c(0,4,0,4,0,4,0), caption = "2-way RMAs for chronic ABG")

```

Table 3: 2-way RMAs for chronic ABG

Metric	Group p-value	*	Day p-value	*	Interaction p-value	*
CtO2	0.0029	**	0.0305	*	0.4283	
Glucose	0.9964		0.0000	***	0.8641	
Hct	0.3818		0.0180	*	0.6556	
Hb	0.7664		0.0198	*	0.4024	
Lactate	0.0022	**	0.0883	.	0.5074	
pCO2	0.0091	**	0.0425	*	0.0225	*
pH	0.7887		0.6615		0.2226	
pO2	0.0033	**	0.8211		0.4391	
sO2	0.0365	*	0.6578		0.4131	

```

abgs |>
  filter(ABGs==TRUE, Study_day%in%(c(0,4,8))) |>
  mutate(Study_day = as.character(Study_day)) %>%
  lme(pCO2 ~ Group + Study_day + Group:Study_day, data = ., random = ~1|ID) -> l_pco2

emmeans(l_pco2, "pairwise" ~ Group:Study_day, adjust = "sidak")$contrasts[c(
  9, # Sham within
  4, # Inf within
  1, # d0 between
  10, # d4 between
  15 # d8 between
)] |> as_tibble() |> kable(digits=c(0,0,3,3,3,3,4), caption="Post-hoc contrasts for PCO2")

```

Table 4: Post-hoc contrasts for PCO2

contrast	estimate	SE	df	t.ratio	p.value
Sham Study_day0 - Sham Study_day8	0	0.834	28	0.293	0.999
Inflate Study_day0 - Inflate Study_day8	-3	0.946	28	-3.639	0.005
Inflate Study_day0 - Sham Study_day0	2	1.486	14	1.292	0.706
Inflate Study_day4 - Sham Study_day4	4	1.486	14	2.841	0.064
Inflate Study_day8 - Sham Study_day8	6	1.486	14	3.773	0.010

Hemodynamics

All metrics analyzed as two-way RMA treating time as a categorical factor. Where appropriate, the following five post-hoc comparisons are examined with Sidak correction: - d0 v d8 within group - d0, d4, d8 between groups

```
hemodynamics |>
  mutate(RPP = HR*(AoP-RAP)) |>
  filter(Allocation=="OHSU", Pressures==TRUE, Study_day%in%(c(0,4,8))) |>
  group_by(ID, Group, Study_day) |>
  select(AoP, RAP, HR, RPP) |>
  pivot_longer(cols=c(4:7), names_to = "Metric", values_to = "Value") |>
  ungroup() |>
  group_by(Group, Metric, Study_day) |>
  summarise(Mean = mean(Value), SD = sd(Value)) |>
  pivot_wider(names_from = Group, values_from = c(Mean, SD)) |>
  relocate(Metric, Study_day, Mean_Sham, SD_Sham, Mean_Inflate, SD_Inflate) |>
  kable(digits = c(0,0,2,2,2,2), caption="Chronic hemodynamic summary")
```

Table 5: Chronic hemodynamic summary

Metric	Study_day	Mean_Sham	SD_Sham	Mean_Inflate	SD_Inflate
AoP	0	42.33	3.90	40.75	2.42
AoP	4	42.73	4.31	40.12	5.29
AoP	8	44.89	6.28	39.82	3.93
HR	0	175.99	15.41	179.46	9.99
HR	4	160.91	12.04	173.35	11.81
HR	8	150.07	13.20	164.91	17.65
RAP	0	3.04	0.88	3.47	1.48
RAP	4	2.66	0.88	4.05	2.28
RAP	8	2.48	0.86	4.42	1.85
RPP	0	6931.47	1072.69	6683.52	324.28
RPP	4	6455.53	877.61	6217.32	796.28
RPP	8	6402.95	1302.18	5769.72	259.37

```
hemodynamics |>
  filter(Pressures==TRUE, Study_day%in%(c(0,4,8))) |>
  mutate(RPP = HR*(AoP-RAP),
         Study_day=as.character(Study_day)) |>
  group_by(ID, Group, Study_day) |>
  select(AoP, RAP, HR, RPP) |>
  pivot_longer(cols=c(4:7), names_to = "Metric", values_to = "Value") |>
  ungroup() |>
  nest_by(Metric) |>
  mutate(mod = list(aov(Value ~ Group * Study_day + Error(ID), data = data))) |>
  summarise(`Group p-value` = summary(mod)[[1]][[1]][1,5], `*` = pstar(`Group p-value`),
           `Day p-value` = summary(mod)[[2]][[1]][1,5], `*` = pstar(`Day p-value`),
           `Interaction p-value` = (summary(mod)[[2]][[1]][2,5]),
           `*` = pstar(`Interaction p-value`)) |>
  relocate(Metric, `Group p-value`, `*`, `Day p-value`, `*`, `Interaction p-value`, `*`) |>
  kable(digits = c(0,4,0,4,0,4,0), caption="Hemodynamic two-way repeated measures ANOVA summary Metric")
```

Table 6: Hemodynamic two-way repeated measures ANOVA summary Metric ~ Group + Study day + Group:Study day

Metric	Group p-value *	Day p-value *	Interaction p-value *
AoP	0.1581	0.3910	0.1455
HR	0.1134	0.0000 ***	0.1120
RAP	0.0519 .	0.9583	0.0983 .
RPP	0.3497	0.0060 **	0.5506

Aortic output: tabular results and statistics

```
hemodynamics |>
  filter(Allocation=="OHSU",Flows==TRUE, Study_day%in%(c(0,4,8))) |>
  group_by(ID,Group, Study_day) |>
  select(AoFl, SV) |>
  pivot_longer(cols=c(4:5), names_to = "Metric", values_to = "Value") |>
  ungroup() |>
  group_by(Group,Metric, Study_day) |>
  summarise(Mean = mean(Value), SD = sd(Value)) |>
  pivot_wider(names_from = Group, values_from = c(Mean,SD)) |>
  relocate(Metric, Study_day,Mean_Sham,SD_Sham, Mean_Inflate,SD_Inflate) |>
  kable(digits = c(0,0,4,4,4,4), caption="Chronic aortic output summary")
```

Table 7: Chronic aortic output summary

Metric	Study_day	Mean_Sham	SD_Sham	Mean_Inflate	SD_Inflate
AoFl	0	388.3840	133.3006	388.9683	101.1895
AoFl	4	427.6794	142.1377	45.1567	109.8427
AoFl	8	455.6154	125.5457	-48.2250	130.9163
SV	0	2.2283	0.8192	2.1764	0.5855
SV	4	2.6494	0.9507	0.2220	0.6375
SV	8	3.0084	0.9416	-0.3438	0.7997

```
hemodynamics |>
  filter(Flows==TRUE, Study_day%in%(c(0,4,8))) |>
  mutate(Study_day = as.character(Study_day)) |>
  group_by(ID,Group, Study_day) |>
  select(AoFl, SV) |>
  pivot_longer(cols=c(4:5), names_to = "Metric", values_to = "Value") |>
  ungroup() |>
  nest_by(Metric) |>
  mutate(mod = list(aov(Value ~ Group * Study_day + Error(ID), data = data))) |>
  summarise(`Group p-value` = summary(mod)[[1]][[1]][1,5], `*` = pstar(`Group p-value`),
    `Day p-value` = summary(mod)[[2]][[1]][1,5], `*` = pstar(`Day p-value`),
    `Interaction p-value` = (summary(mod)[[2]][[1]][2,5]),
    `*` = pstar(`Interaction p-value`)) |>
  relocate(Metric,`Group p-value`,`*`,`Day p-value`,`*`,`Interaction p-value`,`*`) |>
  kable(digits = c(0,4,0,4,0,4,0), caption="Chronic aortic output 2-way RMAs")
```

Table 8: Chronic aortic output 2-way RMAs

Metric	Group p-value	*	Day p-value	*	Interaction p-value	*
AoFl	6e-04	***	0e+00	***	0	***
SV	5e-04	***	6e-04	***	0	***

```
hemodynamics |>
  filter(Flows==TRUE, Study_day%in%(c(0,4,8))) |>
  mutate(Study_day=as.character(Study_day)) %>%
  lme(AoFl ~ Group + Study_day + Group:Study_day, data = ., random = ~1|ID)-> l_AoFl

emmeans(l_AoFl, "pairwise" ~ Group:Study_day, adjust = "sidak")$contrasts[c(
  9, # Sham within
  4, # Inf within
  1, # d0 between
  10, # d4 between
  15 # d8 between
)] |> as_tibble() |> kable(digits=c(0,0,3,3,3,4),caption="Aortic flow post-hoc contrasts")
```

Table 9: Aortic flow post-hoc contrasts

contrast	estimate	SE	df	t.ratio	p.value
Sham Study_day0 - Sham Study_day8	-67	36.021	22	-1.866	0.3241
Inflate Study_day0 - Inflate Study_day8	437	38.907	22	11.237	0.0000
Inflate Study_day0 - Sham Study_day0	1	69.813	11	0.008	1.0000
Inflate Study_day4 - Sham Study_day4	-383	69.813	11	-5.479	0.0010
Inflate Study_day8 - Sham Study_day8	-504	69.813	11	-7.217	0.0001

```
hemodynamics |>
  filter(Flows==TRUE, Study_day%in%(c(0,4,8))) |>
  mutate(Study_day=as.character(Study_day)) %>%
  lme(SV ~ Group + Study_day + Group:Study_day, data = ., random = ~1|ID)-> l_SV

emmeans(l_SV, "pairwise" ~ Group:Study_day, adjust = "sidak")$contrasts[c(
  9, # Sham within
  4, # Inf within
  1, # d0 between
  10, # d4 between
  15 # d8 between
)] |> as_tibble() |> kable(digits=c(0,0,3,3,3,4), caption="Stroke volume post-hoc contrasts")
```

Table 10: Stroke volume post-hoc contrasts

contrast	estimate	SE	df	t.ratio	p.value
Sham Study_day0 - Sham Study_day8	-1	0.242	22	-3.224	0.0194
Inflate Study_day0 - Inflate Study_day8	3	0.261	22	9.643	0.0000
Inflate Study_day0 - Sham Study_day0	0	0.451	11	-0.115	1.0000
Inflate Study_day4 - Sham Study_day4	-2	0.451	11	-5.379	0.0011
Inflate Study_day8 - Sham Study_day8	-3	0.451	11	-7.428	0.0001

Plasma values

Troponin

NOTE: analyzed parametrically using a base-ten log transform.

```
tni |>
  filter(Allocation=="OHSU",ABGs==TRUE) |>
  group_by(ID,Sex,Group) |>
  slice(1) |>
  ungroup() |>
  group_by(Sex,Group) |>
  summarise(n=n()) |>
  pivot_wider(names_from = Sex, values_from = n) |>
  kable(caption="Troponin sample sizes")
```

Table 11: Troponin sample sizes

Group	F	M
Inflate	1	6
Sham	6	3

```
tni |>
  filter(Allocation=="OHSU",ABGs==TRUE) |>
  mutate(logTnI = log10(TnI)) |>
  group_by(Group, Study_day) |>
  select(logTnI) |>
  summarise(Mean = mean(logTnI, na.rm=T), SD = sd(logTnI,na.rm=T)) |>
  pivot_wider(names_from = Group, values_from = c(Mean,SD)) |>
  relocate(Study_day,Mean_Sham,SD_Sham, Mean_Inflate,SD_Inflate) |>
  kable(digits = c(0,0,2,2,2,2), caption="Troponin summary")
```

Table 12: Troponin summary

Study_day	Mean_Sham	SD_Sham	Mean_Inflate	SD_Inflate
0	1	0.32	1.27	0.12
4	1	0.21	1.67	0.21
8	1	0.49	2.25	0.82

```
tni |>
  filter(ABGs==TRUE, Study_day>=0) |>
  mutate(logTnI = log10(TnI)) |>
  mutate(Study_day = as.character(Study_day)) %>%
  lme(logTnI ~ Group + Study_day + Group:Study_day, data = ., random = ~1|ID) -> l_tni

anova(l_tni)
```

```
##              numDF denDF  F-value p-value
## (Intercept)      1    28 355.0263  <.0001
```



```
## Group          1    14 12.4646 0.0033
## Study_day      2    28  4.9364 0.0146
## Group:Study_day 2    28  8.4344 0.0014
```

```
emmeans(l_tni, "pairwise" ~ Group:Study_day, adjust = "sidak")$contrasts[c(
  9, # Sham within
  4, # Inf within
  1, # d0 between
  10, # d4 between
  15 # d8 between
)] |> as_tibble() |> kable(digits=c(0,0,3,3,3,4), caption="Troponin post-hoc contrasts")
```

Table 13: Troponin post-hoc contrasts

contrast	estimate	SE	df	t.ratio	p.value
Sham Study_day0 - Sham Study_day8	0	0.169	28	0.436	0.9958
Inflate Study_day0 - Inflate Study_day8	-1	0.192	28	-5.091	0.0001
Inflate Study_day0 - Sham Study_day0	0	0.212	14	0.053	1.0000
Inflate Study_day4 - Sham Study_day4	1	0.212	14	2.564	0.1076
Inflate Study_day8 - Sham Study_day8	1	0.212	14	4.997	0.0010

ANP

```
hormones |>
  filter(Allocation=="OHSU",ABGs==TRUE) |>
  group_by(ID,Sex,Group) |>
  slice(1) |>
  ungroup() |>
  group_by(Sex,Group) |>
  summarise(n=n()) |>
  pivot_wider(names_from = Sex, values_from = n) |>
  kable(caption="ANP sample sizes")
```

Table 14: ANP sample sizes

Group	F	M
Inflate	1	6
Sham	6	3

```
hormones |>
  filter(Allocation=="OHSU",ABGs==TRUE) |>
  group_by(Group, Study_day) |>
  select(ANP) |>
  summarise(Mean = mean(ANP, na.rm=T), SD = sd(ANP,na.rm=T)) |>
  pivot_wider(names_from = Group, values_from = c(Mean,SD)) |>
  relocate(Study_day,Mean_Sham,SD_Sham, Mean_Inflate,SD_Inflate) |>
  kable(digits = c(0,2,2,2,2,2), caption="ANP summary")
```

Table 15: ANP summary

Study_day	Mean_Sham	SD_Sham	Mean_Inflate	SD_Inflate
0	0.51	0.10	0.62	0.19
4	0.52	0.17	0.63	0.23
8	0.55	0.12	0.69	0.15

```

horomones |>
  filter(ABGs==TRUE, Study_day>=0) |>
  mutate(Study_day = as.character(Study_day)) %>%
  lme(ANP ~ Group + Study_day + Group:Study_day, data = ., random = ~1|ID) -> l_anp

anova(l_anp)

```

```

##              numDF denDF  F-value p-value
## (Intercept)      1    28 315.12513 <.0001
## Group            1    14  3.55861 0.0802
## Study_day        2    28  0.91569 0.4119
## Group:Study_day  2    28  0.13215 0.8768

```

PRA

See ANP for sample sizes.

```

horomones |>
  filter(Allocation=="OHSU",ABGs==TRUE) |>
  group_by(Group, Study_day) |>
  select(PRA) |>
  summarise(Mean = mean(PRA, na.rm=T), SD = sd(PRA,na.rm=T)) |>
  pivot_wider(names_from = Group, values_from = c(Mean,SD)) |>
  relocate(Study_day,Mean_Sham,SD_Sham, Mean_Inflate,SD_Inflate) |>
  kable(digits = c(0,2,2,2,2,2), caption = "PRA summary")

```

Table 16: PRA summary

Study_day	Mean_Sham	SD_Sham	Mean_Inflate	SD_Inflate
0	9.15	8.81	13.61	13.62
4	15.06	8.47	24.30	14.12
8	18.05	13.88	15.48	12.10

```

horomones |>
  filter(ABGs==TRUE, Study_day>=0) |>
  mutate(Study_day = as.character(Study_day)) %>%
  lme(PRA ~ Group + Study_day + Group:Study_day, data = ., random = ~1|ID) -> l_pra

anova(l_pra)

```

```

##              numDF denDF  F-value p-value

```

## (Intercept)	1	28	51.62548	<.0001
## Group	1	14	0.70959	0.4137
## Study_day	2	28	2.83100	0.0759
## Group:Study_day	2	28	1.43730	0.2546

Terminal study

Strain

```
ste |>
  filter(Exclude==FALSE, STE==TRUE) |>
  group_by(ID,Group,Metric,Ventricle) |>
  summarise(Value = mean(Value)) |>
  mutate(Metric = case_when(
    Metric=="Circ" ~ "GCS",
    Metric=="Long" ~ "GLS"
  )) -> STEmeans

STEmeans |>
  group_by(ID,Group) |>
  slice(1) |>
  select(ID) |>
  ungroup() |>
  inner_join(weights) |>
  group_by(Group,Sex) |>
  summarise(n=n()) |>
  pivot_wider(names_from = Sex, values_from = n) |>
  kable(caption = "STE summary")
```

Table 17: STE summary

Group	F	M
Inflate	1	6
Sham	6	2

Strain: tabular results and statistics

```
STEmeans |>
  group_by(Group,Metric, Ventricle) |>
  summarise(Mean = mean(Value), SD = sd(Value)) |>
  pivot_wider(names_from = Group, values_from = c(Mean,SD)) |>
  relocate(Metric, Ventricle, Mean_Sham,SD_Sham, Mean_Inflate,SD_Inflate) ->STESUM

STEmeans |>
  ungroup() |>
  nest_by(Ventricle,Metric)->STENEST

STENEST |>
  mutate(btest = list(bartlett.test(Value ~ Group, data = data))) |>
  summarise(bp = btest$p.value) |>
  mutate(Test = case_when(
    bp > 0.05 ~ "t-test",
    TRUE ~ "U-test"
  )) |>
```

```

full_join(STENEST) |>
unnest(cols = "data") |>
ungroup() |>
nest_by(Ventricle, Metric, Test) |>
mutate(mod = case_when(
  Test=="t-test" ~ list(t.test(Value ~ Group, data = data)),
  TRUE ~ list(wilcox.test(Value ~ Group, data = data))
)) |>
summarise(pvalue = mod$p.value) |>
mutate("*" = pstar(pvalue)) |>
full_join(STESUM) |>
kable(digits = c(0,0,0,4,0,2,2,2,2), caption = "STE summary")

```

Table 18: STE summary

Ventricle	Metric	Test	pvalue	*	Mean_Sham	SD_Sham	Mean_Inflate	SD_Inflate
Left	GCS	t-test	0.9817		-35.73	11.95	-35.94	19.47
Left	GLS	t-test	0.3959		-22.44	9.02	-16.98	13.60
Right	GCS	t-test	0.6954		-34.36	8.95	-32.05	9.01
Right	GLS	U-test	0.9015		-18.25	4.83	-22.66	13.75

Ventricular chamber size / volume

Values derived from STE measurements. Despite desire to match LV and RV measures, Tomtec platform does not make chamber area accessible in LV workflow.

Chamber volume / area tabular summary

```

vols |>
  filter(Exclude==FALSE, STE==TRUE) |>
  group_by(ID, Ventricle, Axis, Metric, Sex, Group) |>
  summarise(Value = mean(Value)) -> vols_MEANS

vols |>
  filter(Ventricle=="Left", Axis == "Long", Exclude==FALSE) |>
  pivot_wider(names_from = "Metric", values_from = "Value") |>
  mutate(EF = (100*(ESV/(EDV-ESV)))) |>
  group_by(ID, Group) |>
  summarise(EF=mean(EF))-> LVEF

vols |>
  filter(Metric%in%c("EDA", "ESA"), Exclude==FALSE) |>
  pivot_wider(names_from = "Metric", values_from = "Value") |>
  mutate(FAC = (100*(EDA/(EDA-ESA)))) |>
  group_by(ID, Group, Ventricle, Axis) |>
  summarise(FAC=mean(FAC))->FAC

LVEF |>
  mutate(Metric = "EF",

```

```

      Ventricle = "Left",
      Axis = "Long") |>
  rename(Value = EF) ->dEF

FAC |>
  mutate(Metric = "FAC") |>
  rename(Value = FAC) |>
  bind_rows(dEF) -> dFACEF

vols_MEANS |>
  ungroup() |>
  select(-Sex) |>
  bind_rows(dFACEF) |>
  filter(!is.na(Value), !is.infinite(Value)) -> vma

vma |>
  group_by(Group, Metric, Ventricle, Axis) |>
  summarise(Mean = mean(Value, na.rm=T), SD = sd(Value, na.rm=T)) |>
  pivot_wider(names_from = Group, values_from = c(Mean, SD)) |>
  relocate(Metric, Ventricle, Axis, Mean_Sham, SD_Sham, Mean_Inflate, SD_Inflate) -> VOLSUM

vma |>
  ungroup() |>
  nest_by(Ventricle, Metric, Axis) -> VOLNEST

VOLNEST |>
  mutate(btest = list(bartlett.test(Value ~ Group, data = data))) |>
  summarise(bp = btest$p.value) |>
  mutate(Test = case_when(
    bp > 0.05 ~ "t-test",
    TRUE ~ "U-test"
  )) |>
  full_join(VOLNEST) |>
  unnest(cols = "data") |>
  ungroup() |>
  nest_by(Ventricle, Axis, Metric, Test) |>
  mutate(mod = case_when(
    Test == "t-test" ~ list(t.test(Value ~ Group, data = data)),
    TRUE ~ list(wilcox.test(Value ~ Group, data = data))
  )) |>
  summarise(pvalue = mod$p.value) |>
  mutate("*" = pstar(pvalue)) |>
  full_join(VOLSUM) |>
  kable(digits = c(0,0,0,0,4,0,2,2,2,2), caption = "Echo geometry summary")

```

Table 19: Echo geometry summary

Ventricle	Axis	Metric	Test	pvalue	*	Mean_Sham	SD_Sham	Mean_Inflate	SD_Inflate
Left	Long	EDV	t-test	0.0275	*	1.16	0.72	0.42	0.35
Left	Long	EF	t-test	0.3730		28.42	22.97	45.17	34.55
Left	Long	ESV	t-test	0.1985		0.23	0.23	0.10	0.10
Left	Short	EDA	t-test	0.0254	*	1.34	0.48	0.64	0.35

Ventricle	Axis	Metric	Test	pvalue	*	Mean_Sham	SD_Sham	Mean_Inflate	SD_Inflate
Left	Short	ESA	t-test	0.0335	*	0.54	0.29	0.16	0.20
Left	Short	FAC	t-test	0.1604		156.18	65.59	117.94	32.62
Right	Long	EDA	t-test	0.0687	.	1.55	0.71	2.19	0.38
Right	Long	ESA	t-test	0.9800		0.67	0.38	0.67	0.49
Right	Long	FAC	t-test	0.5014		156.51	71.55	132.47	76.60
Right	Short	EDA	t-test	0.1981		1.75	0.87	2.33	0.50
Right	Short	ESA	t-test	0.3622		0.73	0.52	1.02	0.46
Right	Short	FAC	t-test	0.8770		145.01	56.05	139.55	70.49

Myocardial perfusion (CEUS)

Sample size summary applies to *all* MCE data:

```
MCE_means |>
  group_by(ID, Group, Ventricle, Study) |>
  select(ABV, Beta, ABF_mL_min_g, O2_mL_min_RPP, O2_mL_min_g, o2ct) |>
  pivot_longer(cols=c(5:10), names_to = "Metric", values_to = "Value") |>
  ungroup() |>
  group_by(Group, Metric, Ventricle, Study) |>
  summarise(Mean = mean(Value), SD = sd(Value)) |>
  pivot_wider(names_from = Group, values_from = c(Mean, SD)) |>
  relocate(Metric, Ventricle, Study, Mean_Sham, SD_Sham, Mean_Inflate, SD_Inflate) -> MCESUM
```

```
MCE_means |>
  group_by(ID, Group, Ventricle, Study) |>
  select(ABV, Beta, ABF_mL_min_g, O2_mL_min_RPP, O2_mL_min_g, o2ct) |>
  pivot_longer(cols=c(5:10), names_to = "Metric", values_to = "Value") |>
  ungroup() |>
  nest_by(Metric, Ventricle, Study) -> MCENEST
```

```
MCENEST |>
  mutate(btest = list(bartlett.test(Value ~ Group, data = data))) |>
  summarise(bp = btest$p.value) |>
  mutate(Test = case_when(
    bp > 0.05 ~ "t-test",
    TRUE ~ "U-test"
  )) |>
  full_join(MCENEST) |>
  unnest(cols = "data") |>
  ungroup() |>
  nest_by(Metric, Test, Ventricle, Study) |>
  mutate(mod = case_when(
    Test == "t-test" ~ list(t.test(Value ~ Group, data = data)),
    TRUE ~ list(wilcox.test(Value ~ Group, data = data))
  )) |>
  summarise(pvalue = mod$p.value) |>
  mutate("*" = pstar(pvalue)) |>
  full_join(MCESUM) -> MCE_SUMMARY_TABLE
```

```
MCE_means |>
```

```

group_by(ID,Group) |>
slice(1) |>
select(ID) |>
ungroup() |>
inner_join(weights) |>
group_by(Group,Sex) |>
summarise(n=n()) |>
pivot_wider(names_from = Sex, values_from = n) |>
kable()

```

Group	F	M
Inflate	1	4
Sham	5	1

```

MCE_SUMMARY_TABLE |>
mutate(
  Metric=case_when(
    Metric=="ABF_mL_min_g" ~ "MBF",
    Metric=="ABV" ~ "MBV",
    Metric=="O2_mL_min_RPP" ~ "Oxygen RPP",
    Metric=="O2_mL_min_g" ~ "Oxygen del",
    TRUE ~ Metric
  )
)|> kable(digits = c(0,0,0,0,4,0,3,3,3,3), caption = "Comprehensive CEUS summary")

```

Table 21: Comprehensive CEUS summary

Metric	Test	Ventricle	Study	pvalue	*	Mean_Sham	SD_Sham	Mean_Inflate	SD_Inflate
MBF	U-test	RV	ADO	0.0079	**	10.501	5.553	63.509	26.009
MBF	t-test	LV	BL	0.0097	**	4.348	2.157	13.032	4.591
MBF	t-test	LV	ADO	0.0439	*	14.534	5.802	27.315	9.999
MBF	t-test	RV	BL	0.0111	*	8.840	7.245	49.443	21.683
MBV	t-test	LV	BL	0.0281	*	0.292	0.115	0.551	0.178
MBV	t-test	LV	ADO	0.5514		0.452	0.139	0.492	0.068
MBV	t-test	RV	BL	0.0150	*	0.305	0.147	0.569	0.119
MBV	t-test	RV	ADO	0.1386		0.342	0.165	0.493	0.118
Beta	U-test	RV	BL	0.0079	**	0.510	0.193	1.639	0.957
Beta	U-test	RV	ADO	0.0079	**	0.544	0.110	2.214	0.563
Beta	t-test	LV	BL	0.0517	.	0.258	0.090	0.443	0.152
Beta	t-test	LV	ADO	0.0626	.	0.598	0.199	0.978	0.329
Oxygen RPP	t-test	LV	BL	0.0523	.	0.000	0.000	0.000	0.000
Oxygen RPP	t-test	LV	ADO	0.1660		0.000	0.000	0.000	0.000
Oxygen RPP	t-test	RV	BL	0.0196	*	0.000	0.000	0.000	0.000
Oxygen RPP	t-test	RV	ADO	0.1222		0.000	0.000	0.000	0.000
Oxygen del	t-test	LV	BL	0.1907		0.348	0.219	0.769	0.587

Metric	Test	Ventricle	Study	pvalue	*	Mean_Sham	SD_Sham	Mean_Inflate	SD_Inflate
Oxygen del	t-test	LV	ADO	0.2398		0.972	0.573	1.509	0.779
Oxygen del	t-test	RV	BL	0.0219	*	0.591	0.704	2.017	0.862
Oxygen del	t-test	RV	ADO	0.0349	*	0.648	0.510	2.668	1.480
o2ct	t-test	LV	BL	0.1133		8.033	2.662	5.440	2.241
o2ct	t-test	LV	ADO	0.4447		6.800	2.845	5.580	2.213
o2ct	t-test	RV	BL	0.2202		5.700	1.920	4.260	1.450
o2ct	t-test	RV	ADO	0.2202		5.700	1.920	4.260	1.450

Reserve

```

MCE_means |>
  select(-ABV,-Beta,-O2_mL_min_RPP, -AoP, -JV, -HR, -RPP, -anesth_time, -O2_mL_min_g) |>
  pivot_wider(names_from = "Study", values_from = ABF_mL_min_g) |>
  filter(!is.na(ADO)) |>
  mutate(reserve = ADO/BL) -> Reserve

Reserve |>
  group_by(Group, Ventricle) |>
  summarise(Mean = mean(reserve,na.rm=T), SD = sd(reserve,na.rm=T)) |>
  pivot_wider(names_from = Group, values_from = c(Mean,SD)) |>
  relocate(Ventricle, Mean_Sham,SD_Sham, Mean_Inflate,SD_Inflate) ->RESERVESUM

Reserve |>
  ungroup() |>
  nest_by(Ventricle)->RESERVENEST

RESERVENEST |>
  mutate(btest = list(bartlett.test(reserve ~ Group, data = data))) |>
  summarise(bp = btest$p.value) |>
  mutate(Test = case_when(
    bp > 0.05 ~ "t-test",
    TRUE ~ "U-test"
  )) |>
  full_join(RESERVENEST) |>
  unnest(cols = "data") |>
  ungroup() |>
  nest_by(Ventricle,Test) |>
  mutate(mod = case_when(
    Test=="t-test" ~ list(t.test(reserve ~ Group, data = data)),
    TRUE ~ list(wilcox.test(reserve ~ Group, data = data))
  )) |>
  summarise(pvalue = mod$p.value) |>
  mutate("*" = pstar(pvalue)) |>
  full_join(RESERVESUM) |>
  kable(digits = c(0,0,4,0,2,2,2,2), caption = "CEUS reserve summary")

```

Table 22: CEUS reserve summary

Ventricle	Test	pvalue	*	Mean_Sham	SD_Sham	Mean_Inflate	SD_Inflate
LV	t-test	0.5254		2.89	1.10	2.41	0.95
RV	t-test	0.7255		1.43	0.78	1.30	0.26

Anesthetized hemodynamics and blood characteristics

Anesthetized ABGs

```
iso_abg |>
  filter(ABGs==TRUE, STE==TRUE, Sample=="Pre") |>
  group_by(ID,Sex,Group) |>
  select(pH,sO2,pCO2,pO2,Hb,Hct,CtO2,Glucose,Lactate,Protein) |>
  pivot_longer(cols=c(4:13), names_to = "Metric", values_to = "Value") |>
  ungroup() |>
  group_by(Group,Metric) |>
  summarise(Mean = mean(Value,na.rm=T), SD = sd(Value,na.rm=T)) |>
  pivot_wider(names_from = Group, values_from = c(Mean,SD)) |>
  relocate(Metric, Mean_Sham,SD_Sham, Mean_Inflate,SD_Inflate) ->IABGSUM

iso_abg |>
  filter(ABGs==TRUE, STE==TRUE, Sample=="Pre") |>
  group_by(ID,Sex,Group) |>
  select(pH,sO2,pCO2,pO2,Hb,Hct,CtO2,Glucose,Lactate,Protein) |>
  pivot_longer(cols=c(4:13), names_to = "Metric", values_to = "Value") |>
  ungroup() |>
  nest_by(Metric) -> IABGNEST

IABGNEST |>
  mutate(btest = list(bartlett.test(Value ~ Group, data = data))) |>
  summarise(bp = btest$p.value) |>
  mutate(Test = case_when(
    bp > 0.05 ~ "t-test",
    TRUE ~ "U-test"
  )) |>
  full_join(IABGNEST) |>
  unnest(cols = "data") |>
  ungroup() |>
  nest_by(Metric,Test) |>
  mutate(mod = case_when(
    Test=="t-test" ~ list(t.test(Value ~ Group, data = data)),
    TRUE ~ list(wilcox.test(Value ~ Group, data = data))
  )) |>
  summarise(pvalue = mod$p.value) |>
  mutate("*" = pstar(pvalue)) |>
  full_join(IABGSUM) |>
  kable(digits = c(0,0,4,0,2,2,2,2), caption = "Anesthetized ABG summary")
```

Table 23: Anesthetized ABG summary

Metric	Test	pvalue	*	Mean_Sham	SD_Sham	Mean_Inflate	SD_Inflate
CtO2	t-test	0.2513		7.39	2.78	6.00	1.57
Glucose	t-test	0.5867		1.59	0.51	1.81	0.95
HCt	t-test	0.6382		38.00	7.76	36.50	3.80
Hb	t-test	0.5445		12.11	2.99	11.36	1.54
Lactate	U-test	0.3240		3.26	1.98	3.64	0.68
Protein	t-test	0.4486		3.78	0.81	3.48	0.58
pCO2	t-test	0.9813		78.62	14.53	78.49	7.18
pH	t-test	0.5586		7.15	0.10	7.18	0.06
pO2	t-test	0.0148	*	26.05	4.34	21.00	2.12
sO2	t-test	0.3571		45.06	14.61	39.11	9.12

Anesthetized hemodynamics: pre-study

```

iso_hemodynamics |>
  mutate(RPP = HR*(AoP-RAP)) |>
  filter(Pressures==TRUE, STE==TRUE, Flows==TRUE) |>
  group_by(ID,Group) |>
  select(AoP,RAP,HR,AoF,SV, RPP) |>
  pivot_longer(cols=c(3:8), names_to = "Metric", values_to = "Value") |>
  ungroup() |>
  group_by(Group,Metric) |>
  summarise(Mean = mean(Value,na.rm=T), SD = sd(Value,na.rm=T)) |>
  pivot_wider(names_from = Group, values_from = c(Mean,SD)) |>
  relocate(Metric, Mean_Sham,SD_Sham, Mean_Inflate,SD_Inflate) ->IHEMOSUM

iso_hemodynamics |>
  mutate(RPP = HR*(AoP-RAP)) |>
  filter(Pressures==TRUE, STE==TRUE, Flows==TRUE) |>
  group_by(ID,Group) |>
  select(AoP,RAP,HR,AoF,SV, RPP) |>
  pivot_longer(cols=c(3:8), names_to = "Metric", values_to = "Value") |>
  ungroup() |>
  nest_by(Metric) -> IHEMONEST

IHEMONEST|>
  mutate(btest = list(bartlett.test(Value ~ Group, data = data))) |>
  summarise(bp = btest$p.value) |>
  mutate(Test = case_when(
    bp > 0.05 ~ "t-test",
    TRUE ~ "U-test"
  )) |>
  full_join(IHEMONEST) |>
  unnest(cols = "data") |>
  ungroup() |>
  nest_by(Metric,Test) |>
  mutate(mod = case_when(
    Test=="t-test" ~ list(t.test(Value ~ Group, data = data)),
    TRUE ~ list(wilcox.test(Value ~ Group, data = data))
  ))

```

```

)) |>
summarise(pvalue = mod$p.value) |>
mutate("*" = pstar(pvalue)) |>
full_join(IHEMOSUM) |>
kable(digits = c(0,0,4,0,2,2,2,2), caption = "Anesthetized pre-study hemodynamic summary")

```

Table 24: Anesthetized pre-study hemodynamic summary

Metric	Test	pvalue	*	Mean_Sham	SD_Sham	Mean_Inflate	SD_Inflate
AoF	t-test	0.0009	***	530.64	166.26	-74.33	163.07
AoP	t-test	0.7942		42.11	9.14	40.76	6.67
HR	t-test	0.2408		180.23	24.54	165.12	12.82
RAP	t-test	0.2311		2.05	5.62	5.27	1.56
RPP	t-test	0.2126		7205.22	1792.37	5878.89	1296.65
SV	t-test	0.0017	**	2.99	0.97	-0.46	1.05

Anesthetized hemodynamics: during study

```

MCE_means |>
  group_by(Group, Ventricle, Study, ID) |>
  select(AoP, JV, HR, RPP) |>
  pivot_longer(cols = c(5:8), names_to = "Metric", values_to = "Value") |>
  ungroup() |>
  group_by(Group, Ventricle, Study, Metric) |>
  summarise(Mean = mean(Value), SD = sd(Value)) |>
  pivot_wider(names_from = Group, values_from = c(Mean, SD)) |>
  relocate(Ventricle, Study, Metric, Mean_Inflate, SD_Inflate, Mean_Sham, SD_Sham) -> MCEHEMOSUM

MCE_means |>
  group_by(Group, Ventricle, Study, ID) |>
  select(AoP, JV, HR, RPP) |>
  pivot_longer(cols = c(5:8), names_to = "Metric", values_to = "Value") |>
  ungroup() |>
  nest_by(Ventricle, Study, Metric) -> MCEHEMONEST

MCEHEMONEST|>
  mutate(btest = list(bartlett.test(Value ~ Group, data = data))) |>
  summarise(bp = btest$p.value) |>
  mutate(Test = case_when(
    bp > 0.05 ~ "t-test",
    TRUE ~ "U-test"
  )) |>
  full_join(MCEHEMONEST) |>
  unnest(cols = "data") |>
  ungroup() |>
  nest_by(Metric, Test, Ventricle, Study) |>
  mutate(mod = case_when(
    Test=="t-test" ~ list(t.test(Value ~ Group, data = data)),
    TRUE ~ list(wilcox.test(Value ~ Group, data = data))
  )) |>

```

```

summarise(pvalue = mod$p.value) |>
mutate("*" = pstar(pvalue)) |>
full_join(MCEHEMOSUM) |>
kable(digits = c(0,0,4,0,4,2,2,2), caption = "Anesthetized CEUS hemodynamic summary")

```

Table 25: Anesthetized CEUS hemodynamic summary

Metric	Test	Ventricle	Study	pvalue	*	Mean_Inflate	SD_Inflate	Mean_Sham	SD_Sham
AoP	t-test	LV	BL	0.2831		38.20	8.88	45	12
AoP	t-test	LV	ADO	0.9615		34.07	7.79	34	8
AoP	t-test	RV	BL	0.7620		41.64	9.14	44	13
AoP	t-test	RV	ADO	0.0601	.	45.08	3.04	35	9
HR	U-test	LV	ADO	0.5368		143.99	30.25	162	8
HR	t-test	LV	BL	0.2872		145.70	21.65	158	10
HR	t-test	RV	BL	0.3119		152.36	25.19	166	9
HR	t-test	RV	ADO	0.3567		158.87	15.07	167	10
JV	t-test	LV	BL	0.0890	.	8.03	2.04	6	2
JV	t-test	LV	ADO	0.6887		10.99	3.36	10	5
JV	t-test	RV	BL	0.2072		8.58	3.04	4	6
JV	t-test	RV	ADO	0.5345		10.42	2.64	8	6
RPP	t-test	LV	BL	0.1232		4426.87	1643.49	6246	1903
RPP	t-test	LV	ADO	0.6508		3473.70	1622.92	3950	1744
RPP	t-test	RV	BL	0.3940		5203.87	2029.26	6547	2630
RPP	t-test	RV	ADO	0.2711		5511.93	677.27	4330	2021

Postmortem morphometry

Morphometry: tabular results and statistics

```
weights |>
  filter(Allocation=="OHSU",Weights==TRUE) |>
  group_by(ID,Sex,Group) |>
  select(Body_kg, Heart_body_g_kg, Heart_g, LA_fixed, RA_fixed, LV_fixed, RV_fixed, Septum_fixed,LV_RV_ratio) |>
  pivot_longer(cols=c(4:12), names_to = "Metric", values_to = "Value") |>
  ungroup() |>
  group_by(Group,Metric) |>
  summarise(Mean = mean(Value), SD = sd(Value)) |>
  pivot_wider(names_from = Group, values_from = c(Mean,SD)) |>
  relocate(Metric,Mean_Sham,SD_Sham, Mean_Inflate,SD_Inflate) ->WTSUM

weights |>
  filter(Allocation=="OHSU",Weights==TRUE) |>
  group_by(ID,Sex,Group) |>
  select(Body_kg, Heart_body_g_kg, Heart_g, LA_fixed, RA_fixed, LV_fixed, RV_fixed, Septum_fixed,LV_RV_ratio) |>
  pivot_longer(cols=c(4:12), names_to = "Metric", values_to = "Value") |>
  ungroup() |>
  nest_by(Metric) -> WTNEST

WTNEST |>
  mutate(btest = list(bartlett.test(Value ~ Group, data = data))) |>
  summarise(bp = btest$p.value) |>
  mutate(Test = case_when(
    bp > 0.05 ~ "t-test",
    TRUE ~ "U-test"
  )) |>
  full_join(WTNEST) |>
  unnest(cols = "data") |>
  ungroup() |>
  nest_by(Metric,Test) |>
  mutate(mod = case_when(
    Test=="t-test" ~ list(t.test(Value ~ Group, data = data)),
    TRUE ~ list(wilcox.test(Value ~ Group, data = data))
  )) |>
  summarise(pvalue = mod$p.value) |>
  mutate("*" = pstar(pvalue)) |>
  full_join(WTSUM) |>
  kable(digits = c(2,2,4,4,2,2,2), caption = "Postmortem weight summary")
```

Table 26: Postmortem weight summary

Metric	Test	pvalue	*	Mean_Sham	SD_Sham	Mean_Inflate	SD_Inflate
Body_kg	t-test	0.6821		4.79	0.55	4.95	0.87
Heart_body_g_kg	t-test	0.8544		6.34	0.50	6.41	0.93
Heart_g	t-test	0.6544		30.30	3.63	31.75	7.59
LA_fixed	t-test	0.0384	*	1.99	0.61	3.30	1.29
LV_RV_ratio	U-test	0.0002	***	1.05	0.17	0.78	0.07

Metric	Test	pvalue	*	Mean_Sham	SD_Sham	Mean_Inflate	SD_Inflate
LV_fixed	t-test	0.0157	*	8.86	1.62	6.86	1.29
RA_fixed	t-test	0.0324	*	1.67	0.36	2.41	0.69
RV_fixed	t-test	0.6912		8.57	1.66	8.96	2.11
Septum_fixed	t-test	0.4562		6.74	1.49	6.25	1.03

Geometry

```

geometry |>
  filter(Allocation=="OHSU",Weights==TRUE) |>
  group_by(ID,Sex,Group) |>
  select(RV_radius_cm, RV_wall_cm, LV_radius_cm, LV_wall_cm, Septal_thick_cm) |>
  pivot_longer(cols=c(4:8), names_to = "Metric", values_to = "Value") |>
  ungroup() |>
  group_by(Group,Metric) |>
  summarise(Mean = mean(Value), SD = sd(Value)) |>
  pivot_wider(names_from = Group, values_from = c(Mean,SD)) |>
  relocate(Metric,Mean_Sham,SD_Sham, Mean_Inflate,SD_Inflate) ->GSUM

geometry |>
  filter(Allocation=="OHSU",Weights==TRUE) |>
  group_by(ID,Sex,Group) |>
  select(RV_radius_cm, RV_wall_cm, LV_radius_cm, LV_wall_cm, Septal_thick_cm) |>
  pivot_longer(cols=c(4:8), names_to = "Metric", values_to = "Value") |>
  ungroup() |>
  nest_by(Metric) -> GNEST

GNEST |>
  mutate(btest = list(bartlett.test(Value ~ Group, data = data))) |>
  summarise(bp = btest$p.value) |>
  mutate(Test = case_when(
    bp > 0.05 ~ "t-test",
    TRUE ~ "U-test"
  )) |>
  full_join(GNEST) |>
  unnest(cols = "data") |>
  ungroup() |>
  nest_by(Metric,Test) |>
  mutate(mod = case_when(
    Test=="t-test" ~ list(t.test(Value ~ Group, data = data)),
    TRUE ~ list(wilcox.test(Value ~ Group, data = data))
  )) |>
  summarise(pvalue = mod$p.value) |>
  mutate("*" = pstar(pvalue)) |>
  full_join(GSUM) |>
  kable(digits = c(2,2,4,4,2,2,2,2), caption = "Postmortem ventricular geometry summary")

```

Table 27: Postmortem ventricular geometry summary

Metric	Test	pvalue	*	Mean_Sham	SD_Sham	Mean_Inflate	SD_Inflate
LV_radius_cm	t-test	0.3353		1.07	0.17	0.98	0.15
LV_wall_cm	t-test	0.8790		0.50	0.06	0.49	0.09
RV_radius_cm	t-test	0.8059		1.35	0.15	1.33	0.19
RV_wall_cm	t-test	0.9468		0.49	0.05	0.49	0.07
Septal_thick_cm	t-test	0.3620		0.62	0.12	0.68	0.12

Growth curves

Using fixed weights, control d8 LV weighed 8.86

Per Jonker's 2015 paper, LV weight prior to 143 dGA is linear at 0.1425 g/day. Therefore, projected LV weight at study d0 is 7.72. Observed experimental LV weight is less than that, at 6.8628571.

As a sanity check, Jonker's analysis suggests that sham hearts at 135 dGA would be $0.1425 \times (135dPC) - 12.03 = 7.2075$